

# Understanding Variables and the Variable Cascade in DDI-CDI

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## Introduction

This paper provides a brief introduction to variables in the Data Documentation Initiative – Cross Domain Integration (DDI-CDI) specification, particularly for those who may not be familiar with the DDI or UN/ECE Modern Statistics metadata models and specifications. Variables are a core construct in many of today’s important metadata models, and their most sophisticated expression may well be the “variable cascade” as seen in the DDI-CDI specification. The variable cascade exists to help us distinguish among the various properties we must have in order to find, understand, and process variables for different functions across the data lifecycle, and particularly in the comparison and integration of data.

DDI-CDI represents the result of many years of thinking and implementation of metadata standards in the social, behavioral, and economic (SBE) sciences, and in the world of official statistics. Notably, the modelling of the variable is the product of more than two decades of evolution, starting with the development of the DDI Codebook specification, published originally in 2000, and ISO/IEC 11179 Metadata Registry standard, originally published between the years 1994 and 2000 in several parts. In these specifications, models for variables and data elements (respectively) were put forward. Subsequent specifications have further developed these ideas, notably the DDI Lifecycle specification (2008) and the Generic Statistical Information Model (GSIM), first published in 2012. The DDI-CDI specification was released at the end of 2024.

It is notable that the evolution of these specifications reflects the demand across the worlds of science and statistics to include more data of more different types in any type of research, whether for policy or academic ends. They have increasingly supported a wider range of data types than were historically deemed important in traditional data management in the domains from which they came. We have also seen the emergence of the FAIR Data Principles and advances in informatics which are indicative of the shift away from narrowly focused domain practice to the sharing of data for different purposes across domain and institutional boundaries. The advent of generative artificial intelligence only reinforces a trend which was already underway: researchers can now

use truly vast amounts of data in an increasingly broad range of methods. The growth in the demand for data is profound.

This trend must be supported by the models used to manage that data: we must be able to describe it from the point of measurement or creation through the activities of data management and dissemination. The specifications noted above have been focused on these challenges for many years, and represent some of the most mature thinking in these areas. The integration of such mature models with the latest technological approaches is critical in the face of this ever-increasing demand for data.

It is hoped that this brief introduction will make one of the core ideas of these specifications – what is termed the “variable cascade” - more approachable for those working in research and informatics who are faced with the current challenges around effectively providing data to researchers and their systems. This model has been implemented in different ways in different specifications, and we anticipate that it can also help to inform further standardization efforts to meet current needs.

## What is a “Variable”?

The word “variable” potentially has a lot of different meanings, even when we look only at its use as a noun. For our purposes, however, we can be fairly narrow in how we understand it. The Cambridge Dictionary gives us the following (<https://dictionary.cambridge.org/us/dictionary/english/variable>):

*a number, amount, or situation that can change:*

- *The variables in the equation are X, Y, and Z.*
- *The data was analyzed according to neighborhoods, but other key variables like credit rating and marital status were ignored altogether.*

This definition gets us quite close to what we mean in research when we describe a variable: it is a field in the data which has a value in it, which is part of each record in the data. The value will change from record to record. For the purposes of analysis the variable can be used in a formula and populated with the set of values found in the data. These values are often numeric, but may also be coded items taken from a controlled list or classification, or consist of other types of constrained values such as time stamps. Such values are typical of what is termed “quantitative data.” The definition above hints at all of this - so far so good.

Data professionals have a larger range of different uses for variables, however, because they are responsible for the data across its entire lifecycle. Variables are defined so they can then be used in a data collection activity and populated; data sets are managed, with changes to the description of a variable – and its meaning for the purposes of

analysis – being versioned as they change. Variables may be reused in repeat collections of the same data, allowing them to be compared effectively with prior measurements. Variables may be the focus of searches for data within a collection of on a network, supporting the location of data. They may drive the processing of data, to make it more accurate during data cleaning or to protect the respondents who provided it. The list goes on and on.

This view of a variable goes well beyond the definition provided by the Cambridge Dictionary, but it helps us to realize an important point. A “variable” is the changeable value, but it is also the *description of that value*. When we talk about variables in the context of statistical metadata, we are more often than not actually discussing the ways in which we describe the variables. It is easy to split hairs here, but what is important is to recognize that although variables themselves may be quite simple, the sufficient description of them to support all the tasks performed by data professionals is distinctly not. (The next section looks at this issue in greater detail.)

We should note that in different domains, the same general idea may be referred to with a different name. In the ISO/OEC 11179 standard, and in many domains, the term “data element” is used, and this is in many senses the same as our variable. In some scientific domains you have the idea of a ‘qualified unit of measure,’ and this is – especially in its more complex form – essentially the same thing as a variable. Some domains which use surveys to collect data will think of variables as synonymous with the questions used to populate them. There are probably many other terms which are used in similar ways.

DDI-CDI uses the term “variable” in the way that it does because that has come to be the accepted terminology in the SBE sciences, in official statistics, and in many other domains. Further, this term is associated with the same metadata construct in the many specifications on which DDI-CDI builds. There is probably no single term which will make all data professionals happy across all domains – what is important is to understand how the term is used for our purposes here, in the DDI-CDI model.

## How are Descriptions of Variables Used?

This section looks at the different ways in which we must describe our variables in order to support the many different functions we need to perform throughout the data lifecycle. We will look at how variables are described for existing data sets, and for operations on those data, and then on cases where variable descriptions are being applied across data sets, or for those data sets which do not yet exist, or are not yet complete.

## Describing Variables in an Existing Dataset

The classic form of data description to many users is the form which appears in a “data dictionary.” Here, we have a definition of what a variable means, and an indication of what values the variable may have, including meaningful (“substantive”) values, as well as flags or codes to indicate statuses such as missing values (“sentinel values”). The original DDI Codebook specification provided an XML format for describing variables as seen in data dictionaries. This was an improvement on the then-typical documentation as a PDF or Word document, because it allowed the variable descriptions to be processed in a granular way, making them much more useful within data archives and other data dissemination systems where the variable descriptions had to be provided to the data users as documentation.

This distinction – having a machine-readable form of the description, rather than just a human-readable one – was a major step forward, and it is even more important today than it was in 2000, as today there are more different ways that machines can access and use the metadata descriptions.

An important part of having a machine-processable description of the variable for an existing data set was that it was possible to perform automated functions on it without requiring human attention. The classic case here was transformation of data between proprietary formats used by different analysis packages. Thus, a data set described using DDI Codebook would have information about the variables and where they could be found in the data file, so that it could be transformed into SPSS formats, SAS formats, STATA formats, etc. as requested by users, and that this could be done generically for all data sets as an on-demand function. Data archives benefitted from what was a significant efficiency improvement – prior to this the transformations were often done manually. A single, neutral form of the data and metadata could be held in an archive or dissemination system for management, and it could be exposed to users in whatever form they required.

This principle of machine-actionability has a major impact on how we model variables today: it requires that not only is the variable itself described, with its definition and representation, but that its location within the data set, and the role it plays structurally in reference to the data set as a whole, are also described. Thus, a field which acts as an identifier for a record (say, an account identifier, a person’s identifier, or the identifier of a material sample) can be noted as distinct from a field which carries an observed measurement or a descriptor of another measurement in the record such as a status field. These roles are very important when we want to programmatically transform data sets into various different forms for different user applications, as they allow us to understand how records made up of a cluster of variables are grouped and identified.

## Describing Variables across Multiple Datasets

There is still, however, a major failing in having an after-the-fact description of a variable as it appears in a single data set: it is not reusable. This has important implications for how useful the variable description actually is. Having “reusable” variables is important in cases where you wish to compare and/or merge data coming from multiple sources – you need to know that the variable encountered in two places is actually functionally the same thing.

At the same time that DDI Codebook was providing machine-actionable descriptions of existing variables, within the context of an extant data set, ISO/OEC 11179 established the idea of a reusable “data element”. We can think of a data element as a template on which a variable is based: it tells you the name and definition of the variable, and describes how it will be represented. This data element is then published in a registry where it is visible to those who are producing data sets. Any given variable can make reference to the data element on which it is based, so that users – both human and machine - can understand that it is the same as other variables using the same template in other data sets.

Later version of the DDI specification, as well as the GSIM model, included this idea of a data element, and then made it an integral part of a single coherent model of variables, leading eventually to the variable cascade which we will describe more fully below.

There are two basic scenarios in which comparable variables appear: cases where data sets are designed to be comparable, to demonstrate changes across time or some other relevant axis, or cases where two data sets have similar variables, and we wish to understand if they are sufficiently alike to be compared for the purposes of whatever analysis we are conducting. These cases are termed, respectively, “comparability by design” and “ad hoc comparability.”

Comparability by design is most often seen in cases where a single entity is collecting or producing data on an ongoing basis. This is the case in many research areas where large institutions produce data for the use of a large population of researchers.

Longitudinal studies are common on some sciences, as are repeat cross-sectional ones. Time series – sets of identical measurements across time on a repeated basis – are typical of official statistics and in many other areas of data production. In order to efficiently manage data production, organizations which produce such data sets often manage their data elements – reused variables – in centralized systems.

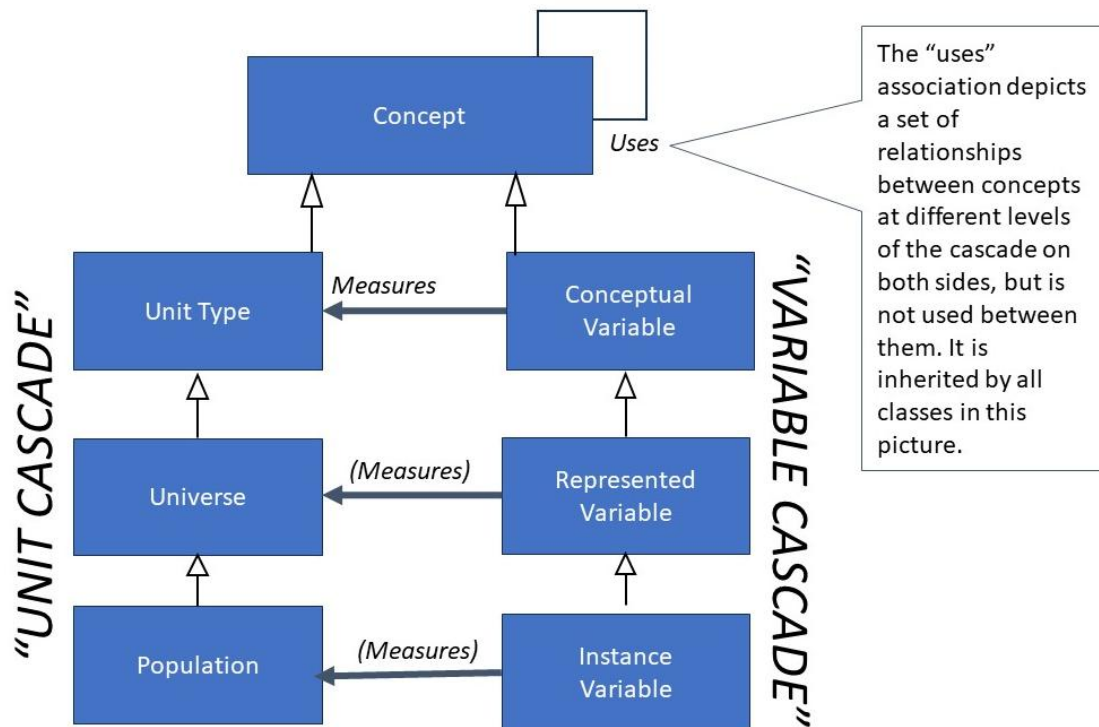
Ad hoc comparability is more typical of scenarios where data is coming from disparate sources or even different domains. This scenario is increasingly common in the era of FAIR data, where the possibility of reusing data is becoming more of a reality. It has

always been present in many research areas, however. Often, it is not a question of finding identical measurements, but of understanding how two similar variables can be processed to produce a value which *is* comparable. This process involves – at least conceptually – the transformation of two different data sets into a single, larger data set, and is often termed “data integration.” The feasibility and quality of the resulting data is often determined by the specifics of the common variables which are driving the integration, and having a detailed knowledge of such variables is extremely important. Variables are comparable to a greater or lesser extent in such scenarios, and it is important to understand the fashion and degree to which they are the same and different.

What we can know about the reusable and comparable aspects of variables is not identical to what we can know about a single variable in the context of a single data set. The comparable information is a subset of the full description of the variable in context. This aspect of variable description is what has given rise to the nuanced model which we call the “variable cascade.”

## The Variable Cascade

“Variable cascade” is a term which is shorthand for the section of the DDI-CDI model – and similar models in specifications from the DDI Alliance and elsewhere – which looks like the right-hand column in the diagram below:



It is worthwhile walking through the classes illustrated in this diagram both in the variable cascade and the parallel "unit cascade", even though that is a secondary consideration for the discussion here, as the Unit, Universe, and Population are often not recorded in the metadata in practice.

We start with a Concept. This is a term and its formal definition. If you are familiar with the SKOS specification, its idea of a concept very much aligns with what is found in that specification. Such a term/definition pairing can be used for a wide variety of purposes.

A Conceptual Variable is a Concept being used to refer to and define a property of some object. A simple example is "age". This is a concept that applies to many different types of real-world things: people have ages, buildings have ages, geological features have ages, and so on. When used to provide a term and definition for a variable – that is, as a Conceptual Variable – the concept age must be applied to the thing for which it is a property. This thing – the object – is termed a Unit in DDI-CDI (think "unit of interest" not "unit of measurement").

A Conceptual Variable has all the properties of a Concept, but it adds to these a link to its Unit. Further, it may qualify some aspects of the Concept – "age" may become "person age", etc. because of this, the Conceptual Variable inherits all of the properties of a Concept in the DDI-CDI model. Conceptual Variables are often referred to in cases where an ad hoc comparison is being performed – two variables have the same

definition but not the same type of values (i.e., age can be expressed as an integer, or can be expressed as a code for a cohort such as “5 to 10 years”, etc.)

The Represented Variable is a Conceptual Variable to which has been added information about how the values of the variable will appear in the data. Thus, a variable might hold a time stamp, or a code referring to a node in a classification or controlled vocabulary, or it might be some numeric type, etc. A Represented Variable inherits all the properties of a Conceptual Variable. The Represented variable is the type of variable most commonly reused for the purposes of comparison by design.

Represented Variables typically apply to a constrained group of the potential members of all possible Units, which in DDI-CDI is termed a “Universe”. For example, if a Represented Variable measures the age of teenagers, using integers from 12 to 19, that is a subset of the more generic Unit “persons”, for which such a restricted representation would not be useful.

Representations are often the result of the nature of the Unit being observed, For persons, integers in full years and age cohorts might be very common representations for an age variable; for geological features it might be more typical to give the age based on a geological time scale.

The Instance Variable is the fully contextualized description of a variable as it appears in a single data set. It is possible to say things about an Instance Variable which cannot be generically described for a Represented variable, because the details are dependent on the data set in which the variable appears. A common example of this is variable-level statistics: how often was a particular valid value used in the data set? If I have a variable which has a binary “true-false” value, I might want to record that it has a “true” value in 65% of the cases in my data set. Such information cannot be described for variables except in the context of a data set. For this reason, the Instance Variable class inherits all the properties of the Represented Variable, but adds some additional details of this type.

An Instance Variable is used to hold the values for a known set of cases – what is termed a “Population” in DDI-CDI. A population is a subset of a Universe, but it has been completely qualified to reflect the specific set of objects being observed or measured. In many cases, the Population is represented by a Sample, in which case it is this which is actually measured in a data set.

In DDI-CDI, the “uses” relationship exists to provide links between these different classes of variable description: I may wish to maintain my Represented and Conceptual Variables independently from my Instance Variables within a system, especially if I am producing data which is intended to be comparable by design. I would connect them with the “uses” relationship. However, it is also always the case that an Instance

Variable has sufficient information to act as a Represented Variable or as a Conceptual Variable, and a Represented Variable always has sufficient information to act as a Conceptual Variable. Whether it is useful to maintain separate identities for these different flavors of variable description within a system is dependent on the system in question, and what it is trying to achieve.

Simple systems intended to describe data sets as they exist, in isolation, will be able to employ only Instance Variables. Studies producing time series or longitudinal data might wish to persist and manage Represented Variables separately from Instance Variables, and so on. (We see this difference in the way that the variable cascade is implemented in the DDI-CDI specification, and in the DDI Lifecycle specification, because the latter specification is more focused on the cyclical production of official statistics and longitudinal or repeat cross-sectional data. In this specification, Represented Variables are always persisted and maintained with their own identities.)

## Variables in the Description of Data Structures

One of the significant uses of variable description in systems is to enable the automated processing of data sets. In order to perform operations on data sets, it is often necessary to understand what the component variables are, and how they relate to one another. In other words, the structure of the data set needs to be formally described at a sufficient level of detail to support processing.

Examples of this include reshaping data sets from (for example) “long” to “wide,” or the re-packaging of data from a database or data warehouse into a specific tabular form to respond to a specific query.

There are two things which are critical in understanding the logic content of the data: the set of variables which make up the available records about any specific case, and the roles which can be played by those variables in different structures.

In DDI-CDI (and in some other DDI specifications), there is a construct termed a Logical Record, which is the set of variables available for any specific case. (The Logical Record can be bounded by a data set or by a repository, depending on what is needed.) The roles which can be played by variables are determined according to how the data is structured within the data set or repository. There are four fundamental patterns for structuring data: as “wide” data, where each record is presented as a row in a table; as “long” data where each measurement value is presented as a row in a table, along with an identifying key; as key-value pairs within a pool of data (e.g., a data warehouse); or as a matrix (or “cube”) with dimensionalized keys identifying each observation value or measurement.

The DDI-CDI model describes the many different roles variables can play in these different structural types, and shows how they change when the structure of the data set they are in is changed. A field which is an identifier in a wide data set might need to be supplemented with other fields to provide a sufficient key when those same data values are structured in a long data set or a multi-dimensional one (etc.).

Information about the roles played by variables belongs not to the description of the Variables, but to the description of the data structure which references the variable and assigns a role to it. In the DDI-CDI model, this structural function is performed on Represented Variables (or on Instance Variables being treated as Represented Variables in cases where separate Represented Variables are not described and maintained).

## The Value of the Variable Cascade

The variable cascade gives us a nuanced way of describing the set of information we know about a variable, allowing use to determine exactly what we can do with that variable. I cannot, for example, understand how to integrate two data sets if I do not know if the representations of two conceptually equivalent variables is the same: I need agreement at the level of the Represented Variable for such a task. For searches, it may well be that having a description of the Conceptual variables is all that I require. For the purposes of assessing the fitness of a particular data set for an analysis, I may want to see summary statistics about the variable in the context of the data set, and thus require a description at the level of the Instance Variable.

When comparing and integrating data sets, understanding the degree to which different variables correspond requires having a model which can express this level of detail. For building time series and longitudinal views of data, knowing that two variables are based on the same Represented Variable is critical; for ad hoc comparison, having the full descriptive range of Conceptual and Represented Variables becomes useful in understand what is and is not possible through automated transformations. If I need to reuse data across unlike systems, understanding the variables and the roles they play within the structure of their data sets or repositories is critical for the purposes of machine actionability. Ambiguity in what constitutes a variable description can be extremely limiting, whereas a rich and nuanced description of variables can be a powerful enabler.

Especially in an age when we have an increased ability to work across data sets from different sources, and when we have tools such as generative AI which can give us the ability to really make use of disparate data sources, it is important that we have a sufficiently powerful model to describe exactly what our data measures, and how. That is the purpose of the variable cascade.